



FACULTY OF SCIENCE COMPUTER AND TECHNOLOGY AND INFORMATION
DEPARTMENT OF SOFTWARE ENGINEERING AND INFORMATION TECHNOLOGY

SWE 4949A -1: BACHELOR PROJECT

FINAL YEAR PROJECT PROPOSAL

Semester 2, 2025/2026

**QR-BASED HOSPITAL VISITOR AND PASS MANAGEMENT SYSTEM WITH BLOCKCHAIN
TECHNOLOGY**

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1. Introduction

1.1 Background of the Study

Digitalization has become more common in most of the industries, including the healthcare industry. Digitalization fundamentally transformed how medical institutions operate and deliver services. For example, digitalization simplifies the manual visitor management process in conventional hospitals which is one of the main management issues that hospitals often take for granted and ignore. Digitalization of the visitor management system is essential due to the variety of visitor types in hospitals. Every hospital visitor has different purposes and locations to visit. For example, contractors come to conduct daily maintenance tasks, vendors come to deliver medical supplies or equipment, caretakers (penjaga) come to take care of patients and stay for a longer time; executive visitors come to visit patients in a shorter time. Thus, a structured digital management system helps hospitals to perform better visitor management, control and tracking, while enhancing the security, operational efficiency, and institutional accountability of hospitals.

Hospital Sultan Abdul Aziz Shah (HSAAS), UPM is a major, high human flow academic medical center located in Seri Kembangan, Selangor. HSAAS has its own workflow for managing the visitors. During visiting hours, HSAAS allows executive visitors to visit patients without going through the registration and pass issuance process. However, the executive visitors are requested to register and obtain a physical pass when visiting patients out of visiting hours. Meanwhile, other visitor types like vendors, contractors and caretakers are requested to register once they enter the hospital. Passes will be issued based on the location they want to visit. All the passes need to be returned before the visitors depart from the hospital. All these visitor activities are managed manually using a logbook and paper-based form at a pass issuance counter. This seriously increases the complexity of the management of visitors because it must accommodate a diverse population of walk-in visitors across multiple functional categories. Bartenschlager et al. (2022) also highlighted that manual visitor admission processes in hospitals can significantly affect user experience because of the long waiting times while increasing the staff workload, because of the weak monitoring system efficiency. This finding reflects similar operational challenges currently faced by HSAAS.

Healthcare security literature consistently documents the risks associated with unmanaged physical access in the healthcare environment. Inefficient access control mechanisms and real-time monitoring in healthcare environments could significantly affect both digital and physical security. A survey on security and privacy threats in healthcare systems was conducted by Newaz et al. (2021) showing 55.6 percent reported attacks were passive in nature, making them difficult to detect without an active monitoring mechanism. Moreover, López Martínez et al. (2023) highlighted the importance of identification, authentication, authorization, and accountability mechanisms in protecting healthcare environments against a wide range of security threats. Qureshi and Koo (2026) further documented that the average data breach costs reached USD 7.42 million, the highest cost among all industries in 2025. Inadequate physical and logical access controls were identified as the major contributing factors of the loss. This highlights the growing importance of integrating operational monitoring and access management into the healthcare environment.

Manual visitor management using paper is not sufficient anymore to meet the operational and monitoring needs of hospitals nowadays because of its low traceability and accountability. A study on hospital visitor and chaperone tracking during the COVID-19 pandemic conducted by Huang et al. (2022) highlighted that digitalized visitor logging enables accurate contact identification and ward -level traceability that manual logbooks cannot provide. Similarly, Ng and Ong (2022) conducted an experiment at Singapore Hospital to implement a real-time location-aware tracking system in a 1,000- bed tertiary. The result shows that the efficiency of the digital tracking system was three times more efficient compared to conventional manual review methods. The digitalized contact tracking system successfully reduced the required time for contact tracing by 96.2 percent. These findings highlight how efficient and significant digital systems are compared with paper-based systems.

Visitor management using QR code has been validated for adoption in institutional environments. Rao and Jonathan (2019) highlight the feasibility and effectiveness of QR-based solutions by demonstrating a contactless and pre-registered check-in for smart offices using QR code. Similarly, a cloud-integrated visitor management system with features including real-time host notification, visitor movement tracking, and analytics capabilities, proposed by Mohaideen et al. (2025) also further identifies hospitals as a suitable deployment environment for such a

system. However, none of the existing systems fully address the operational workflow and security requirements of HSAAS that relies on the issuance and return tracking of physical visitor passes, the integration of visitor registration with patient Medical Record Numbers (MRN), or the management of physical access credentials used to enter restricted ward areas.

While QR-based visitor management systems provide greater efficiency for register and monitoring operations, they also tend to use traditional, centralized databases to record visitor/pass transactions. These audit logs remain vulnerable to unauthorized modification, deletion and administrative override, reducing the reliability of audit logs as accountability evidence. Ullah et al. (2024) showed that blockchain technology could create audit trails which are tamper-proof and immutable by creating a cryptographically verifiable record of transactions; Stock et al. (2022) also confirmed the feasibility of using blockchain technology in physical space for visitor authentication and access control. The research indicates that blockchain could enhance the traceability, accountability, and trust in the management of visitor passes. To this end, the proposed solution for this project is the integration of blockchain technology as an immutable audit trail layer to record the pass lifecycle events of visitors, such as pass issuance, return, overdue events and administrative changes.

Levina et al. (2024) also articulates that the integration of multiple individual systems such as data analytics and security system into a unified architectural model is essential for modern medical institutions to achieve full efficiency. Therefore, this proposal aims to design and develop a practical and deployable system that digitalizes existing visitor pass counter workflow at HSAAS by introducing a QR-based counter registration for walk in visitors while providing real-time operational visibility through a staff dashboard without requiring visitors to pre-register online.

The development of the proposed system will adopt the Agile Kanban methodology. Kanban supports continuous and incremental development by visualising development tasks, controlling work in progress, and allowing pending work to be reprioritised when stakeholder needs change (Ahmad et al., 2018; dos Santos et al., 2018). This approach is suitable for the project because the system is developed by a single student developer. Continuous involvement from HSAAS will allow completed functions to be reviewed and the Kanban backlog to be

refined throughout development, helping the system remain aligned with actual hospital requirements.

This project proposes the development of QR-based Hospital Visitor and Pass Management with Blockchain Technology for HSAAS. The system aims to digitalize current HSAAS manual visitor and pass management workflow and enhance the auditability of visitor and pass record. A physical QR code is displayed to allow visitors to scan and register through digital forms through their device while the counter staff verify the submission and record physical pass issuance and return using a digital dashboard. The pass lifecycle events are recorded as tamper-resistant blockchain transactions to keep the reliability and integrity of the pass transaction record. Besides, administrators are able to manage passes and users, review the analytic results and audit trails through the dashboard. Overall, the proposed system keeps immutable audit trails, simplifies and digitalizes visitors and pass management workflow, and increases the visibility and traceability of registered visitor pass activities at HSAAS.

1.2 Motivation of the Study

This study was motivated by operational observations of the current visitor management process during a site visit to HSAAS, where no digital recording and tracking system of visitors was observed, and the system of manual logbooks, paper-based registration forms and physical passes, managed by a counter-based system, completely lacked automation. This project is driven by six interrelated realities of operation.

Currently, HSAAS caters for four different types of visitors: Executive Visitors, Penjaga (Caretakers), Vendors and Contractors, with different registration procedures, access rights and processes. For example, Penjaga visitors are required to have a form with the patient's MRN bar code from the ward to register at the pass counter. The existing manual workflows are conducted purely on paper, causing inconsistencies, delays and making it impossible to search, analyze or audit them afterwards. In their study, Lin et al. (2022) showed that the digitization of hospital registration procedures, including walk-in visitors, further enhances the quality of the data, minimizes administrative burden on staff, and allows systematic data management.

Secondly, at HSAAS, physical visitor passes are approximately RM30 each, and are the main passes used to gain entry to specific wards or restricted areas. The hospital at present has a

high pass non return rate, meaning that visitors come and go without taking their passes back to the visitor's counter where their identity cards will be retained. With no digital system to monitor the issuing and return status of each pass, the hospital has no automated means to find out who has an unreturned pass, to remind staff if a pass is out for longer than the allowed time, or to call the visitor with their registration phone number.

Thirdly, the hospital does not currently have a centralized view of hospital activities on different wards and restricted areas. In a systematic review of the literature on healthcare analytics, Bichescu et al. (2025) identified real-time operational visibility as a critical element that aids resource allocation, operational monitoring, and decision-making in healthcare organizations. HSAAS can struggle to identify trends among visitors, monitor the level of occupancy in each ward, and then react to any problems as they occur without a monitoring system. As a result, a real-time registered visitor analytics dashboard would give HSAAS increased operational visibility and give them the data they need to make informed decisions.

Fourthly, current visitor management solutions including digital QR-based systems usually keep audit records in traditional relational databases, which are subject to retroactive changes or deletion by database administration. Ullah et al. (2024) reported that lack of tamper-proof audit trail mechanisms in healthcare access systems can lead to accountability issues and exacerbate security risks. This directly fills the accountability void highlighted by Cobrado et al. (2024) and enhances the compliance posture of HSAAS under Malaysia's PDPA 2010.

Fifthly, beyond operational accountability, Malaysia's updated Personal Data Protection Act (PDPA 2010) places greater responsibility on organizations to manage personal data carefully and protect individual privacy. Hamin et al. (2025) noted that organizations which collect data are expected to actively protect and secure the personal data collected following proper data management practices. Thus, HSAAS as a big organization which frequently interacts with a high volume of personal data, should implement a system which offers a clear and transparent means of gathering, holding and processing visitor data, aiding adherence to PDPA regulations.

Finally, the proposed system adopts the Agile Kanban development methodology. Kanban enables development tasks to be visualised and managed continuously through a Kanban board.

Stakeholder feedback can be incorporated by adding, refining, or reprioritising pending work items without waiting for the completion of a fixed development sprint. This supports the evolving operational requirements of HSAAS and allows the software to be progressively refined throughout the development lifecycle.

1.3 Project Overview

The project proposes a Hospital Visitor and Pass Management System with Blockchain Technology for HSAAS to address the current operational problems in visitor registration and physical pass management. Development will use Agile Kanban, in which requirements and development tasks move continuously through the stages of Backlog, Ready, In Progress, Review and Testing, Stakeholder Validation, and Done. Completed functions will be reviewed with HSAAS and subsequent work will be reprioritised when necessary.

The system mainly involves three types of users: administrator, counter staff, and visitors. The administrator is responsible for counter staff and pass management, analytics reporting, and blockchain audit trail review. Visitors can scan a QR code to access and submit the registration form without creating a system account. Different visitor categories are provided with forms based on their required information and access purposes. Counter staff verify submitted visitor information before issuing physical passes and manage the pass lifecycle activities through the dashboard.

The proposed workflow begins when a walk-in visitor scans a QR code displayed at the hospital pass counter to access a digital registration form through the visitor's own device web browser without installing any application or creating any user account. The digital form requests visitors to select relevant visitor categories, namely Executive Visitor, Penjaga, Vendor, or Contractor. Each visitor category has its own specific registration form to be filled in. For example, Penjaga visitors are required to enter the patient's MRN, which will be validated against the hospital patient information system API.

After the registration submission, the registration record appears on the counter staff dashboard for verification. Counter staff verifies the registration, then assigns appropriate physical pass and records the pass issuance event in the system. When the visitor leaves the hospital, visitors need to return the pass and counter staff record the return of the physical pass

through the dashboard. If a pass is not returned within the configured maximum permitted duration, the system generates an overdue alert to support counter staff follow up and monitor unreturned passes more effectively. All pass lifecycle events including pass issuance, return, overdue, administration modification are recorded through blockchain audit trail module to strengthen the integrity and reliability of pass transaction records.

Based on the proposed workflow, there are several main operational components needed to support a structured and working system. Firstly, the data analytics component which provides an analytical dashboard and historical reporting for visitor and pass activities. Then, the second component is blockchain-anchored audit trails components which records selected critical pass lifecycle events as verifiable blockchain references while the last component is management system components which allows administrators and counter staff to manage visitor records, pass inventory, counter staff accounts, destination information, and system parameters.

The scope of the proposed system is limited to visitors who are required to register and obtain a physical visitor pass through the hospital pass counter, particularly during non-visiting hours or special access situations. General visitors during normal visiting hours who are not required to obtain a physical pass remain outside the scope of this system.

2. Problem Statement

2.1 Problem Identification

HSAAS currently faces several operational challenges in managing hospital visitors. These challenges were identified through direct observation and discussion conducted during industry site visits. These challenges include:

1. Manual visitor registration process:

HSAAS visitors currently rely on handwritten logbooks and paper forms to do registration at the pass issuance counter. Different types of visitors have different registration procedures that need to be followed and different passes will be assigned after the registration. Among the visitors, the patient's caretaker has the lengthiest workflow because he/she needs to obtain an additional form containing the patient's Medical Record Number (MRN) from different counters before registration can be completed. The conventional

workflow is time-consuming, difficult to manage and does not provide a convenient mechanism for searching and retrieving visitor records when needed.

2. Management of physical visitor passes:

The passes function as access credentials that allow entry into wards and restricted areas. However, The current process relies fully on manual monitoring. Counter staff do not have a centralized digital system to monitor and record the pass issuances. As a result, the counter staff fails to identify and recover the passes when the visitor forgot to return the passes, leading to both financial losses and accountability problems. HSAAS tries to collect identity cards (IC) from the visitor before the pass issuance and request them to collect back their IC before departing to solve the problem. However, HSAAS found the solution was not effective.

3. Lack of real-time operational visibility:

HSAAS currently has no centralized system that provides a real-time overview of registered pass holders and visitor pass across wards and restricted areas. Information such as the number of active visitors, visitor distribution, visitor categories, and pass utilization status is not readily available.

4. Visitor data governance and traceability:

Since HSAAS uses paper forms and logbooks to record visitor information, the historical records are difficult to search, analyze and retrieve. Visitor information is important for HSAAS to generate visitor statistics and maintain a comprehensive audit trail for administrative and compliance purposes.

2.2 Problem Justification

The problems identified are not limited to HSAAS alone but to the healthcare institutions. The related problems have been widely discussed in the existing literature from several approaches such as operational efficiency, security, access control and accountability, and smart hospital and analytic perspectives. Thus, solving these problems is required in order to assist HSAAS in continuing its digital transformation and smart hospital programs.

1. Operational Efficiency

HSAAS has currently an onerous paper based visitor registration system with a great deal of manual paper work. Manual management requires a high administrative effort, lack of data traceability, and long processing time. Manual admission systems are mentioned that result in delayed response times and extra administrative workload, which is discussed by Bartenschlager et al. (2022). The digital registration systems are more efficient and less time consuming. Likewise, Huang et al. (2022) noted that the digital visitor logging system is more accurate and efficient in monitoring visitors than the paper-based system, and Ng and Ong (2022) reported that digital systems can significantly enhance the identification and monitoring of visitors.

2. Security, access control and accountability

Healthcare security literature consistently emphasizes the importance of access control, accountability and monitoring mechanisms. Newaz et al. (2021) highlights that weak access control mechanisms and weak monitoring systems bring significant impact to healthcare institutions especially from information security and privacy aspects. Similarly, López Martínez et al. (2023) emphasized that missing a structured security mechanism in the healthcare system causes the system to continue to face a wide range of security threats. Qureshi and Koo (2026) also documented that data breaches in healthcare institutions suffer the highest loss across all the industry sectors, averaging USD 7.42 million in 2025, Their findings also highlight the necessity of access control, continuous monitoring, and governance mechanisms to mitigate operational and security risks.

3. Blockchain-Enhanced Audit Trail Mechanisms

Operational audit logs found in common relational databases are helpful for operational queries, but fail to offer cryptographic guarantees of data integrity. Any administrator with adequate database permission can add/remove historical data without leaving any trace. To address these issues, Ullah et al. (2024) showed how blockchain technology can provide an immutable audit trail, which transparently documents the details of the transactions, such as the transaction hash, digital signature, and time stamp, and cannot be altered without compromising the cryptographic integrity of the chain. The feasibility of blockchain-based physical access control systems has been supported in real institutional environments by Stock et al. (2022), which showed that these systems are able to provide an audit without any possibility of tampering with the chain. Therefore, the implementation of blockchain technology within

HSAAS pass lifecycle audit trails can help to strengthen the auditability, integrity, and accountability of visitor pass records.

4. Smart hospital and analytic perspectives

To enhance the visibility and decision making of hospitals, it needs a digital monitoring and analytic system. Levina et al. (2024) defines a smart hospital architecture as an architecture that combines monitoring, access control, and information management capabilities into a single architecture. Bichescu et al. (2025) emphasized that analytics-driven systems play an important role in enhancing operational performance and resources management. Likewise, Talib et al. (2023) confirms that real-time monitoring and automated alerts in the healthcare system are feasible to improve its efficiency. These studies reinforce the need for HSAAS to have a system framework that incorporates a real-time monitoring system, an access control and analytics dashboard for greater visibility and control of operations, administrative decision making, and improving the efficiency of visitors at HSAAS.

2.3 Impact of the Problem

The problems identified pose a major impact on HSAAS from an operational, security, auditability and data management perspective.

In an operational sense, using manual visitor registration adds more work to the counter staff and slows down the visitor admission process. Staff must manually check visitors' details, give out physical passes and record visitors details in logbooks/paper. This can lead to delays, possible errors in data entry, and delays in processing visitors. Furthermore, paper records are not easily synchronised and are hard to access when historical information is required. These restrictions can lead to low productivity, inefficient visitor management, and difficulties.

Physical visitor passes are issued as an access card to enter wards and restricted areas within the hospital from a security and accountability point of view. If there's no organized monitoring system, HSAAS could end up with a problem in identifying who has the present pass at hand, if a pass has been returned, and who is still in controlled areas. This harms accountability and can result in the loss of a pass, misuse of a pass or access to a sensitive hospital area. In addition, lost or returned passes may lead to loss of finances as the cost of

replacing the physical pass is about RM30. If this happens often, the total cost and security risk could pose a problem to hospital operations.

Conventional paper records may not be reliable sources of evidence that the record of pass issuance, pass return, overdue passes, and record of pass modification hasn't changed. If the pass lifecycle is not anchored to a blockchain, key pass lifecycle information could be missing cryptographic evidence of integrity. While these are a good way to store and retrieve data in an operational setting, these traditional databases can be open to users with enough system or database privileges to modify the data in a way that may be unauthorized or retroactively done. This can diminish the effectiveness of institutional accountability and affect visitor pass record reliability.

HSAAS has weaknesses in effectively managing visitor data from a data management and compliance standpoint, based on the current paper-based system. It is hard to search, synchronize, analyze, and retrieve visitor records for administrative review and investigation or reporting. There may also be risks of incomplete records, lost forms, inconsistent method of data input and record management issues if visitor information is kept in paper records.

A lack of formal digital documentation could pose difficulties for institutions when it comes to adhering to institutional policies and data protection standards. Without structured electronic documentation, the role of data governance and accountability becomes increasingly important, and without it, it could pose problems in maintaining adherence to institutional policies and data protection standards.

3. Objectives

The following goals are used for the development of the proposed system:

The first is to design and implement a module for walk-in visitor registration that can be done using QR for four types of visitors, namely Executive Visitor, Penjaga (Caretaker), Vendor and Contractor. This objective will be met by allowing the visitors to register via the digital registration form on the browser and have the registration data show up on the counter staff dashboard for additional verification.

The second goal is to build a module for managing the lifecycle of the visitor passes, which will log pass issuance as well as pass return, pass overdue and pass status changes. The overall purpose of this module is to ensure that full transaction records are maintained during the life of a visitor pass and improved accountability over the physical provision of visitor passes.

The third goal is to incorporate blockchain technology as a tamper-evident audit trail mechanism for a subset of the visitor passes transactions. The transactions involved in this are pass issuance, pass return, pass overdue events, and administrative changes. The purpose of this objective is to strengthen the integrity and verifiability of important pass lifecycle records.

Fourth, build a real-time monitoring and analytics dashboard to bring visibility to the operational status of the visitor passes, the distribution of visitors, the status of the visitor pass and the out-of-date pass statistics. This goal helps hospital administrators and counter staff to make better decisions, report and monitor.

4. Scope

4.1 System Scope

The proposed system emphasizes on the management of the visitors who need to register and get physical visitor passes through the hospital pass counter HSAAS. The system is not to be used for the general admission to the premises during the normal hours of the day, when there is no registration or pass process. It instead concentrates on the visitors who are needed to be formally registered, have special access, access outside visiting hours or access to controlled hospital areas.

The system has 5 modules. The first module is the Visitor Registration Module. This feature enables the visitors to scan a QR code at the HSAAS pass counter to take part in a category-specific digital form on their own device. The categories of visitors supported include Executive Visitor, Penjaga (Caretaker), Vendor and Contractor. The system requires the Medical Record Number (MRN) of the patient to be submitted for verification purposes when the penjaga visits the patient, which will be done if the hospital information system has an API integration.

The second module is the Visitor Pass Management Module. This module enables counter staff to check the information of the visitor which is submitted to them on their arrival

before issuing a physical visitor pass. The system captures pass issuance, return, status changes and late pass events during the entire visitor pass lifecycle. This allows hospitals to track if a pass is available, issued, over-due, returned, lost, or disabled.

The Blockchain Audit Trail Module is the third module. This module brings in the blockchain technology to provide verifiable audit records of selected critical pass transaction events. These events are pass issue, pass return, overdue pass and administration change. Each selected event is related to a blockchain transaction reference, which can include a cryptographic hash reference, timestamp, event type and pass reference ID. Sensitive personal information, including details about visitors, contact information and patient information, will be kept in the local database but not entered directly into the blockchain network. To maintain privacy while ensuring integrity and accountability of pass transaction records, only transaction metadata and hash references are stored on-chain.

The Monitoring and Analytics Dashboard Module is the fourth module. This module has been designed to give real-time monitoring of active visitor passes, overdue pass alerts, pass inventory status, visitor distribution, visitor category statistics and historical visitor activity trends. Staff and administrators can easily see pass activities on visitors' dashboards and identify potential issues more quickly.

The Administration Module is the fifth module. The module allows authorized administrators to manage user accounts, configure system settings, manage pass inventory, review audit logs, and generate reports. Administrators may also verify blockchain transaction records by checking the blockchain transaction ID associated with each pass lifecycle event.

4.2 User Scope

The proposed system has three major users: Visitor, Counter Staff, and Administrator.

The visitor is responsible for scanning the QR Code, filling in the digital visitor's registration form and submitting the necessary visitor information according to the type of visitor. There is no system account to create and visitors are not allowed to use administrative functions.

Counter Staff are to check and confirm the visitor information submitted, issue and receive physical visitor passes, update the visitor pass status, attend to visitor pass overdue, and act on alerts in the system. The main operational users of the system at the hospital pass counter are the Counter Staff.

Administrators are responsible for user management, pass inventory management, system configuration, report generation, analytics review, audit trail monitoring and blockchain transaction verification. Administrators are provided with institutional accountability, system maintenance and operational supervision management level functions.

4.3 Dataset Scope

The dataset used by the proposed system consists of visitor registration information, visitor pass transaction records, audit logs, and blockchain transaction references generated during system operation.

Data Category	Data Fields	Storage Location
Visitor Registration Data	Name, IC Number, Phone, Category, Ward, Date/Time, Patient MRN (for Penjaga visitors)	Local Database
Visitor Pass Data	Pass ID, Pass Status, Issue Time, Return Time, Overdue Status, Assigned Staff	Local Database
Audit Trail Data	User Actions, Login Activities, Pass Management Activities, Administrative Changes, Timestamps	Local Database
Blockchain Transaction Data	Blockchain Tx ID, Event Type, Timestamp, Pass Reference ID, Transaction Hash	Blockchain Network (On-chain)

5. Literature Review

5.1 Introduction of Literature Review

This part aims to examine the existing research across six major topics which are related to the proposed QR-Based Hospital Visitor and Pass Management System with Blockchain Technology: (1) security and access control in healthcare environments; (2) hospital visitor management and digital admission processes; (3) QR code-based registration and digital workflow systems; (4) smart hospital infrastructure, operational analytics, and data governance; and (5) visitor traceability, blockchain audit trails and data governance in hospital settings; and (6) Agile software development methodology in healthcare information systems. Together, these themes establish the theoretical foundation for the proposed system and identify the specific operational and research gaps that the proposed solution is designed to address. The review integrates empirical evidence, systematic review findings, and architectural frameworks from thirty peer-reviewed sources.

5.2 Related Works / Previous Studies

5.2.1 Security and Access Control in Healthcare Environment

In a security and privacy threat survey in healthcare systems conducted by Newaz et al., they classified attacks by the nature and the system that they affected, and found that 55.6 percent of the attacks documented were passive in nature, such as eavesdropping, credential exploitation, and impersonation. They are passive attacks; that means they are not triggered or alert any existing controls, and thus are not easily detected. This study highlights the insufficient security managements opens an attack surface to a wide range of security and privacy threats. A foundation countermeasure is to implement role based access control (RBAC) in conjunction with real-time monitoring to solve the problem. The recommended solution is to enhance access monitoring and access accountability efficiency by utilizing RBAC and visitor pass transaction logging, based on the study.

López Martínez et al. (2023) analyzed the cybersecurity threats in healthcare by mapping the documented threats to MITRE ATT&CK framework. The study shed light on the growing complexity of healthcare systems and the need for security measures like identification,

authentication, authorization, and accountability to ensure the safety of healthcare operations. Additionally, the authors pointed out that healthcare organizations remain a very appealing target for any number of security attacks because of the fact that they are dependent upon a series of interconnected technologies and sensitive information. The results highlight the need for organized mechanisms of controlling and monitoring visitors in hospitals. The discovery encourages the use of actual pass tracking and overdue alarm features in the proposed solutions to enhance accountability and bolster visitor access monitoring.

In addition, Qureshi and Koo (2026) analyzed 52 studies and concluded that multi-layer by integrating access control, audit logging, continuous monitoring, and incident response would be a countermeasure to counter cyber-physical threats in the healthcare environment. Additionally, they also reported that the losses from data breaches in healthcare are at an average of USD 7.42 million, much higher than other industries. Their research also underscores the need for access control, constant surveillance, and governance protocols in healthcare cybersecurity. This directly steers the design pass issuance, tracking and recovery workflow in the proposed system.

Osamika et al. (2025) highlights its importance and efficiency in ensuring the security and reliability of the electronic health records in hospitals. But these two are yet to be played to their full potential in many healthcare settings. They also explained that structured access monitoring mechanisms are important in detecting and reacting to abnormal access patterns to enhance healthcare security. In the same way, Cobrado et al. (2024) confirmed the importance of the Identification, Authentication, Authorization and Accountability (IAAA) in health care access control systems. They also found that there are few mechanisms of accountability within current access control systems. The solution proposed is aimed at filling this gap by introducing a digital access card tracking and monitoring solution especially for physical access control.

A review by de Carvalho Junior and Bandiera-Paiva (2018) discussed the implementation of Role-Based Access Control (RBAC) in healthcare systems. The review highlighted that RBAC is widely implemented in hospital management systems. However, they identified several implementation gaps within current implementations including authorization management, access delegation, emergency access requirements, and the need for more flexible access control mechanisms. Besides, several recommended RBAC practices, such as separation

of concern, permission-role review mechanisms are not consistently implemented. These studies highlight the importance of setting a clear permission boundary when designing a system. The proposed idea is designed to adopt an RBAC model consisting of three clearly defined roles, each limited to only what they need to do as a countermeasure. A **Visitor** can only fill in a registration form. A **Counter Staff** member can view the dashboard and manage passes. Lastly, an **Administrator** can configure the system and access reports. Only necessary access is given instead of excessive access, to perform good RBAC security practices.

5.2.2 Hospital Visitor Management and Digital Admission Processes

Bartenschlager et al. (2022) conducted a one-week experiment to evaluate 4 different hospital visitor admission configurations at a 1000 – bed German University Hospital, with one hundred iterations performed for each admission configuration. The result showed that the average time required for completing the admission processes using QR-assisted configuration was only 9.59 minutes, which was 89.49 percent shorter than manual registration which requires an average of 70.16 minutes. Nevertheless, the maximum waiting queue length also dropped from 173 people to 84 people. The experiment supports that digital registration significantly reduces the staff workload by simplifying the visitor admission process. Besides, the research also highlights the potential of digital systems to enforce stricter visiting regulations and support hospital visitor real-time tracking. This finding motivates the implementation of real-time hospital visitor tracking and visitor duration monitoring features in the proposed solution.

Huang et al. (2022) investigated the operational and epidemiological implications of tracking hospital visitors and chaperones during the COVID-19 pandemic, emphasizing that a structured digital visitor logging system is more accurate, responsive and efficient in identifying contact compared with manual logbooks. The studies show a digital medical automated system responded quickly and decreased total hospitalization cases about 50 percent to free manpower and space during the pandemic. In addition, they also highlighted the necessity of digital records in supporting routine operational management and faster responses to health incidents in hospitals. These studies support the need to design a digital record system which captures complete visitor identity details to ensure better traceability, searchability, and accessibility of digital record of every visit.

5.2.3 QR Code-Based Registration and Digital Workflow Systems

Rao and Jonathan (2019) highlighted the feasibility of QR code-based visitor registration system in real institutional settings through a practical demonstration. The demonstration supports that contactless, browser-accessible QR check-in significantly reduces the front-desk workload and improves processing efficiency. Similarly, later research from Mohaideen et al. (2025) further extended the QR code-based registration to a cloud. This study further identified hospitals where need secure operational monitoring and reliable visitor real-time information tracking as one of the best environments to implement QR code-based registration systems. Both studies validate the feasibility of adopting QR code-based registration system in hospital environment. However, neither study addressed the specific operational requirements of HSAAS, such as physical pass lifecycle management, MRN-based patient linkage for caretaker visitors, or overdue pass detection.

Zulkifli, N. D. M., et al. (2023) created and tested a smart management waiting system for outpatient clinics in a Malaysian hospital. The system utilizes QR code for patients to scan and identify themselves and automatically assigns them a queue number through their mobile phone. Their finding supported the willingness of Malaysia hospital patient to adopt to QR code system because of its effectiveness and convenience. Besides, the integration of live administration dashboard with a QR code-based system is also essential to track patients efficiently, especially during peak operating hours. Furthermore, Arumugam et al. (2025) reviewed existing IoT-based tracking systems in public hospitals concluded that the processing times drop significantly while the experience for both staff and visitors improves when hospital digitalized the management system. Together, these studies strengthen the argument that implementing a digital visitor and pass management system equipped with a live administrative monitoring dashboard at HSAAS is the right move.

5.2.4 Smart Hospital Infrastructure, Operational Analytics, And Data Governance

Levina et al. (2024) referred TOGAF enterprise architecture and created a smart hospital architecture framework. The framework integrates multiple layers including business processes, information systems, data management and technology infrastructure layers into a unified model.

The study highlighted that multiple layers architecture should be integrated by hospitals for identification, access control, monitoring, and information exchange to improve operational efficiency and healthcare service delivery. The proposed solution refers to the four-layer architecture when doing design and development. The business architecture represents the visitor pass counter workflow in HSAAS, while the application architecture includes the frontend, backend and dashboard system. Moreover, the data architecture manages visitor records, analytic data and audit trail storage. Lastly, the technology architecture covers the hosting infrastructure, QR code display system, browser-based visitor forms and staff dashboard platform.

Bichescu et al. (2025) reviewed 414 healthcare operations management studies, emphasizing that hospitals rely on analytics-driven systems to improve operational performance, resource utilization, and decision-making processes. Similarly, a cloud-based smart hospital monitoring system which supports real-time monitoring and automation alert generation demonstrated by Talib et al. (2023). The system features improve operational responsiveness through continuous data collection and immediate notification. Furthermore, Salins et al (2025). studied and reviewed an IoT-based hospital wayfinding system. 85.2 percent of participants considered the system easy to use while the navigation time reduced 87 percent and improved hospital visiting experience. These studies motivate the integration of digital monitoring, analytic dashboard and automation alert system in HSAAS to improve operational visibility, administrative efficiency and visitor experience.

5.2.5 Visitor Traceability, Blockchain Audit Trails and Data Governance in Hospital Settings

Ng and Ong (2022) demonstrated the feasibility of real time location system (RTLS) based digital tracking system at a 1000-bed tertiary hospital in Singapore. The demonstration supports the system traced nearly three times more contact than the traditional methods which review the electronic medical records (EMR) manually. The required contact tracing time reduces 96.2 percent while the manpower hours decrease 97.5 percent. The study highlighted that structured digital tracking systems are more reliable and faster in monitoring the contact in hospital environments. However, the proposed solution does not implement a full RTLS. The proposed system records visitor data including arrival time, destination and departure time which aims to achieve more systematic visitor management purpose and replace traditional manual

logbook in hospital. Similarly, Wang et al. (2023) demonstrated a digital contact tracing system. The system centralized visitor registry and collected health information. The demonstration emphasizes the importance of a centralized database in a hospital environment to monitor disease outbreak and make response more effective.

Furthermore, Goh et al. (2025) highlighted the benefits of healthcare institution adopting electronic medical record (EMR) systems such as enhancing patient coordination, tracking and information accessibility. The proposed idea borrows this same idea of connecting systems together but applies it to visitors. This is a similar concept that sums up the proposed idea to connect systems to visitors. It establishes a structured, traceable relationship between the visitor record and the patient's information, by using the patient's Medical Record Number (MRN), thus enhancing visitor monitoring, administrative verification and subsequent contact tracing within the hospital or health facility.

Ullah et al. (2024) showed that blockchain technology is capable of offering a strong tool for maintaining unalterable audit trails in healthcare information systems. The authors pointed out that these gaps in accountability, and thus security vulnerabilities, are due to the lack of effective audit trail systems. The results of these findings can contribute to the development of the proposed ideas by incorporating a blockchain-based audit trail module, which will improve traceability, accountability, and the integrity of the process of visitor pass transactions.

Stock et al. (2022) showed that the technical feasibility of AIO-based blockchain-based physical visitor authentication and access control systems can be achieved through smart contracts and the Internet of Things (IoT). Their results demonstrate the potential of blockchain to effectively and securely log physical access events. This is to support the use of blockchain technology in the implementation of SHVPMACS to record transactions of the life cycles of visitors passes and ensure access accountability.

In addition, Avina and Sinha (2022) also emphasized that integration of automated alert mechanisms and rule-based monitoring capabilities can significantly improve incident monitoring efficiency. Their work inspires the incorporation of rule-based monitoring and notification mechanisms in proposed solutions to help counter staff to deal with visitors and passes handed to the visitors.

Moreover, Suzumoto et al. (2024) highlights that audit logs in healthcare information system are valuable. These logs can be used to provide valuable information such as a pattern of visitors, peak visitor times, and system activity. These insights can be used by hospital administrators to further understand current hospital management and operational issues. Moreover, data patterns can be used as a tool to identify anomalies and to test the system's performance. With the problems identified, hospitals can take targeted improvement initiatives to improve management processes and overall operational efficiency.

Finally, the amendments to Malaysia's Personal Data Protection Act (PDPA) 2010 was discussed by Hamin et al. (2025). They identified proper security and data management practices to minimize vulnerabilities when dealing with important data. These regulatory changes further reinforce the need of the proposed solution to provide role-based access control and controlled data processing practices and configurable data retention mechanisms for the handling of visitor information and patient-related records.

5.2.6 Agile Software Development Methodology in Healthcare Information Systems

Healthcare information systems are developed within complex operational environments in which requirements may change as stakeholders gain a clearer understanding of existing workflows and proposed digital solutions. Traditional sequential development approaches are less responsive once implementation has begun. Agile approaches instead support incremental delivery, stakeholder involvement, frequent feedback, and adaptation to changing operational requirements.

Kokol (2022) reviewed healthcare-related Agile Software Development literature and concluded that Agile approaches can support software quality, resource management, and collaboration between healthcare and software professionals. The review also highlighted the dynamic nature of healthcare software requirements and the value of iterative development for digital transformation initiatives. Alotaibi and Almudhi (2023) similarly identified adaptability, stakeholder engagement, continuous feedback, and monitoring as important practices for Agile adoption in healthcare management.

Kanban is an Agile and Lean-oriented approach that manages development as a continuous flow of work. Work items are visualised on a board, moved through explicitly defined workflow states, and controlled through work-in-progress limits. Ahmad et al. (2018) conducted a systematic mapping study of Kanban in software engineering and identified its use in supporting continuous development, delivery of value, and responsiveness to change.

Dos Santos et al. (2018) synthesised empirical evidence concerning the benefits and challenges of Kanban in software engineering. Their results identified work visibility, control of project activities and tasks, flow of work, and time-to-market as the benefits with the strongest supporting evidence. These findings are relevant to the proposed project because a visible workflow can help the student developer organise system features, testing work, documentation, defects, and stakeholder feedback within one manageable process.

Zulkifli et al. (2025) examined Agile-Scrum in a hospital information system project and reported improvements in teamwork, responsiveness to requirement changes, and project effectiveness. The study demonstrates the value of Agile stakeholder feedback in healthcare development. However, Scrum structures development around fixed-length sprints and prescribed roles such as Scrum Master, Product Owner, and Development Team. These arrangements are more naturally suited to a development team than to this Final Year Project, which is implemented by one student developer.

Agile Kanban is therefore selected while retaining the Agile principles supported by the healthcare literature. Kanban does not require the project to create artificial team roles or wait for a sprint boundary before accepting feedback. Requirements, defects, technical tasks, and stakeholder comments can be recorded as work items in a prioritised Kanban backlog. The developer pulls work into development only when capacity is available, completes the required testing, and obtains stakeholder validation for operationally significant functions.

Ahmad and Wasim (2023) also noted that Agile development in healthcare should balance flexibility with documentation, regulatory compliance, and data protection obligations. For the proposed system, the Kanban workflow will therefore include explicit review and testing activities, documentation requirements, PDPA-related checks, and stakeholder validation before relevant work items are considered completed.

Collectively, the reviewed studies support the use of Agile development in healthcare information systems and provide a basis for applying Agile Kanban to the QR-Based Hospital Visitor and Pass Management System with Blockchain Technology. The proposed workflow will use the stages Backlog, Ready, In Progress, Review and Testing, Stakeholder Validation, and Done. Work-in-progress limits will be used to reduce excessive parallel work and encourage completion before new tasks are started.

System functionality will be delivered progressively through work items covering QR-based registration, visitor pass lifecycle management, overdue alerts, MRN validation, blockchain audit trails, analytics, and administration. Feedback from HSAAS can create new work items or reprioritise pending items at any point in development. Thus, Agile Kanban provides a transparent and manageable development approach that supports continuous improvement and alignment with the operational expectations of HSAAS.

5.3 Theoretical Framework / Models

The proposed system is based on five theoretical models that together constitute a comprehensive basis for the design, implementation and evaluation of the system:

1. Role-Based Access Control (RBAC) Theory
2. Visitor Pass Lifecycle Management Model
3. Blockchain Audit Trail Model
4. Smart Hospital Architecture Framework Based on TOGAF
5. Agile Kanban Development Framework

5.3.1 Role-Based Access Control (RBAC) Theory

Role-Based Access Control (RBAC) is the concept of least privilege applied to the system, where system access is given based on organizational roles instead of user identities. This helps minimize unauthorized access as it only allows access to functions that are needed to perform tasks. The proposed idea intends to build access control for its operational needs based on the Hospital Pass Counter workflow.

The system contains a public interface for visitors to register and two authenticated system roles. Visitors do not need to have a system account. They are able to open the QR code based browser registration form, choose their visitor category and enter information. Counter

Staff are authenticated users that can view visitor submissions, verify and issue physical passes, monitor overdue passes and conduct operational pass management activities. Administrators can have higher level access privileges such as user management, pass inventory management, system configuration, parameter maintenance, analytics reporting, audit log review and blockchain transaction verification.

This role structure adheres to the RBAC principle by having a separate set of roles for public visitor registration and internal hospital roles. This guarantees that important administrative tasks and functions can only be performed by authorized personnel and administrators. The RBAC design follows access control guidelines provided by Newaz et al. (2021), Cobrado et al. (2024), and de Carvalho Junior and Bandiera-Paiva (2018), paying attention to the protection of healthcare information systems and access control management based on user responsibility.

5.3.2 Visitor Pass Lifecycle Management Model

The Visitor Pass Lifecycle Management Model is a conceptual model used to manage physical passes during their use. In this project, a controlled access pass is being used for each visitor to the hospital, and the pass is also a physical asset of the hospital. A pass is a collection of passes with a current status which indicates availability and usage status, and all significant status changes are stored in pass transactions.

The primary pass statuses are: Available, Issued, Overdue, and Lost or Disabled. If a pass is not assigned to a visitor, it starts in the Available state. The pass will be changed to Issued when it is given by counter staff as a result of a valid identification. If the pass is not returned within the specified due period, the system will then set the pass status to Overdue and create an alert that the pass is overdue for counter staff follow up. The system marks a Returned transaction and changes the status of the pass to Available when the visitor returns the pass. In the event that a pass is permanently lost, damaged, or not usable it may be record as Lost or Disabled.

The pass transaction records are updated with a time-stamp and user information on pass lifecycle events such as issuance, return, overdue generation and administrative modification. Some important pass lifecycle events are also recorded via the blockchain audit trail layer, for

tamper-proof verification of critical transactions. The proposed hospital visitor pass management system will benefit the proposed system by improving the accountability, traceability and operational monitoring. It also aligns with the accountability part of the Identification, Authentication, Authorization, and Accountability (IAAA) concept discussed by Cobrado et al. (2024) in regard to physical visitor access management.

5.3.3 Blockchain Audit Trail Model

The Blockchain Audit Trails Model is an overview of how the blockchain can be used as a verification layer to selected events of the visitor pass life cycle. In this project, blockchain is not used to replace the local database. Rather, it's employed to lend integrity and auditability to important pass transactions. The local database still handles the complete visitor registration, pass data, operational data and personal data.

The blockchain audit trail provides on-chain and off-chain data separation. The on-chain data contains the transaction metadata that do not involve in the encryption of the pass, including pass event type, pass event timestamp, pass reference ID, event hash, transaction reference. Personal information such as personal visitors, contact details, identification information, and details about the patient itself are not kept directly on the blockchain. All the visitor registration information and operational information is stored in a local database and is controlled by the system access. All the information regarding the visitor registration and operational record is stored off-chain in a controlled system access local database.

The on-chain and off-chain data are connected by reference to transactions and cryptographic hash. A hash reference is generated whenever a pass lifecycle event is selected as, for example, on the issuance of a pass, return of a pass, generation of an overdue pass, or administrative modification of a pass; the corresponding blockchain transaction reference is then stored. In verification, the system could verify the transaction data on the local database with the reference on the blockchain and find out if there is any change in the transaction data on the local database. This adds a layer of security and accountability, minimizing the possibility of compromising sensitive personal information.

This design considers the principles raised by Ullah et al. (2024) and Stock et al. (2022) that blockchain can be used for an immutable audit record and verify integrity. In the proposed solution, blockchain technology is thus used specifically in order to enhance the accountability of

audit actions in the critical points of the pass life cycle, while preserving privacy, keeping sensitive data within the controlled local database environment.

5.3.4 Smart Hospital Architecture Framework Based on TOGAF

The structure of the overall system follows the Smart Hospital Architecture Framework (SHAF) proposed by Levina et al. (2024), which is derived from the TOGAF Enterprise Architecture Principles. This is helpful because it will align the business processes, application modules, data management, and technology infrastructure of an organization's hospital to a structured system design.

The proposed system can be used for the walk-in registration and issuance of physical passes at the hospital from the Business Architecture viewpoint. It targets visitors who must pick up a pass at the pass counter and receive a physical pass, e.g. during non-visiting hours and/or special access situations.

Application Architecture-wise, the system comprises the QR based Visitor Registration module, Counter Staff Dashboard module, Pass Lifecycle Management module, Administration module, Analytics Dashboard module, and Blockchain Audit Trail module. The modules are designed to be integrated to meet the operating requirements of visitor and pass management.

The system handles structured data from the Data Architecture point of view, including records of visitors, visitor categories, destinations, pass inventory, pass transactions, audit logs, analytics data and blockchain transaction references. Information relevant to visitors, patients is kept in the local database and selected transaction hashes and references are documented via the blockchain audit trail layer.

On the Technology Architecture side, the system features a browser-based QR registration interface, staff and administrator dashboards, backend services, a relational database, and integration with the blockchain testnet. This technology structure is conducive to digital registration and tracking, monitoring, reporting and audit verification of passes.

In general, the smart hospital architecture concept based on the TOGAF model contributes to the design of a smart hospital system that is not only a technical application, but a

system that is not only in harmony with the workflow of the hospital but also with the objectives of the smart hospital and digitalization of the hospital.

5.3.5 Agile Kanban Development Framework

The development methodology of the proposed system is based on the Agile Kanban Development Framework. Kanban manages development through a visual, pull-based workflow rather than fixed-length sprints. Requirements, technical tasks, testing activities, defects, documentation, and stakeholder feedback are represented as work items that move through explicitly defined workflow stages.

The proposed workflow consists of Backlog, Ready, In Progress, Review and Testing, Stakeholder Validation, and Done. Work-in-progress limits will control how many items may be developed or tested simultaneously. A work item will only move to Done after its acceptance criteria, testing, relevant documentation, and stakeholder validation have been completed. This framework supports flexibility, task visibility, controlled development flow, and continuous feedback from HSAAS (Ahmad et al., 2018; dos Santos et al., 2018).

5.4 Research Gap Analysis

There are strong theoretical and empirical foundations from the existing body of literature regarding hospital visitor management, access control systems in healthcare, smart hospital infrastructure, hospital healthcare analytics and Agile software development. Despite these areas, there is still a substantial gap between these fields and their practical application in Malaysian public hospital's visitor access management workflow.

Bartenschlager et al. (2022) showed the operational advantages of digital visitor admission systems. However, it was a pre-registration workflow which has a big difference compared to Malaysia public hospital which consider walk-in visitor as the dominant. Besides, Rao and Jonathan (2019) and Mohaideen et al. (2025) studies support the feasibility of QR based visitor registration system but without implementing the features of physical pass lifecycle management, category specific workflows, visitor-patient linkage using MRN, and the ability to monitor expired passes.

Access control, accountability, audit logging, and continual monitoring are consistently mentioned as important in healthcare settings by Newaz et al. (2021), López Martínez et al.

(2023), Qureshi and Koo (2026), Osamika et al. (2025), and Cobrado et al. (2024). However, existing studies mainly revolve around digital information systems and EHR systems, and few have studied how to practically apply those principles in the area of physical pass management and access control of visitors within a hospital.

Although Bichescu et al. (2025), Talib et al. (2023), and Salins et al. (2025) identified the benefits of using healthcare analytics, monitoring dashboards, and digital operational systems, they didn't really integrate visitor registration, pass lifecycle management, operational monitoring, and analytics reporting into a single hospital workflow. Current analytics studies are more prioritizing patient management, hospital operations or healthcare infrastructure, but not on visitor and physical access management.

Additionally, Kokol (2022) pointed out that even though the advantages of Agile methodologies in managing changing requirements and collaboration with stakeholders in the development of a healthcare information system are known, the methodologies are underutilized. What little research is done, does not necessarily integrate agile development processes with a hospital-specific visitor management system.

While Ullah et al. (2024) and Stock et al. (2022) have separately validated blockchain as an effective immutable audit trail mechanism for healthcare access systems and physical access control systems respectively, no existing published work applies blockchain audit trail technology specifically to hospital visitor pass lifecycle management — the operational context of HSAAS. This project fills that specific gap.

So far there are no published works suggesting a system that meets all the following:

1. Does not require visitor application installation, supports QR-based counter forms that allow visitors to register by walking up to the counter
2. Handles all aspects of the return of physical visitor passes, issuing, overdue reporting, returning and tracking
3. Supports several categories of visitors, each having different registration procedures/flows and MRN linkage to cater for caretaker visitors
4. Delivers a real-time analytics dashboard that can visualize past patterns of visitor flow, ward occupancy, pass utilization, overdue pass data and operational

5. Integrates blockchain technology as an immutable audit trail layer for all pass lifecycle events, providing tamper-proof accountability without storing personal data on-chain.
6. Maintains digital audit trail that is compliant with the PDPA for accountability and compliance
7. Developed using an Agile Kanban process that supports workflow visualisation, controlled work in progress, flexible prioritisation, and continuous stakeholder validation throughout the development lifecycle.

This project aims to fill the composite gap and presents the QR-Based Hospital Visitor and Pass Management System with Blockchain Technology which combines the functionality of visitor management, pass lifecycle management, blockchain-anchored audit trail, access monitoring, analytics reporting, and Agile-driven development into a single solution that is suitable to the HSAAS.

5.5 Summary

The literature reviewed reveals a consensus that the operational, technical and security aspects of hospital visitation need to be digitized. The success of QR-based registration systems, digital visitor tracking tools, audit trail creation, health care analytics and smart hospital designs, along with the role-based access control have been shown in enhancing operational efficiency and monitoring within health care environments.

What the literature also highlighted were issues that are still not resolved: physical visitor pass processing, temporary visitor credential processing, incorporating patient information with visitor information, visitor accountability in physical access events using blockchain technology and operational visibility and analytics-driven decision making in hospitals.

To address these issues, the proposed QR Based Hospital Visitor and Pass Management with Blockchain Technology will include an integrated QR-based visitor registration, physical pass lifecycle management, MRN visitor validation, real-time monitoring of operations, analytics dashboard reporting, and generate an immutable audit trail with blockchain technology, all in one solution, and with role-based access control. The system not only enhances the efficiency of visitor registration and passes accountability but also provides verifiable, tamper-proof evidence

of all pass lifecycle events through blockchain integration to address the accountability gap identified across the literature and advancing HSAAS's smart hospital transformation objectives.

The use of Agile Kanban supports flexible development and responsiveness to stakeholder feedback and changing operational needs throughout the project lifecycle. The Kanban board provides visible control over requirements, implementation, testing, stakeholder validation, and completed work. The QR-Based Hospital Visitor and Pass Management System with Blockchain Technology therefore integrates healthcare security, visitor management, pass lifecycle management, healthcare analytics, smart hospital infrastructure, blockchain auditability, and Agile software development into a context-specific solution for HSAAS. The proposed system improves visitor registration and pass accountability while supporting data-driven decisions through a real-time analytics dashboard.

6 Methodology

6.1 Introduction

This chapter discusses the research methodology and development approach adopted for the design and implementation of the QR-based Hospital Visitor and Pass Management with Blockchain Technology. The chapter includes overall **research design, development methodology, data collection methods, system design, software verification and validation strategy**, and the **evaluation methods** used to assess the quality and effectiveness of the delivered system. The methodology is designed to ensure that the system is aligned with the real operational requirements of Hospital Sultan Adul Aziz Shah (HSAAS) throughout the development lifecycle.

6.2 Research Design

The project adopts the Design and Development Research (DDR) approach as the research design methodology. DDR focuses on the design, development, implementation and evaluation of software solutions to address identified operational problems. The research is conducted through several phases:

Phases:	Objectives
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Phase 1: Needs Analysis (Problem Identification)	Understand the existing visitor and pass management workflow at HSAAS through site visits, direct observations, and discussion with hospital personnel.
Phase 2: Design and Development	Define and identify the functional and non-functional requirements of HSAAS stakeholders. Based on the requirements, the system architecture, database structure, blockchain integration strategy, and user interface designs are designed and developed.
Phase 3: Evaluation and Validation	Functional testing is conducted to ensure that system requirements are correctly implemented, while usability evaluation and user acceptance testing are performed with relevant stakeholders to assess system effectiveness, usability, and suitability for hospital operations

6.3 Development Methodology

The QR-Based Hospital Visitor and Pass Management System with Blockchain Technology will be developed using the Agile Kanban methodology. Kanban is a continuous-flow development approach that visualises development activities, controls work in progress, and enables work items to be prioritised according to stakeholder value, technical dependency, risk, and project needs. Unlike a fixed sprint-based approach, work items can be started and completed continuously according to available development capacity.

Kanban is suitable for this Final Year Project because the system is developed by a single student developer. The project supervisor and industry supervisor provide guidance, while HSAAS staff participate as stakeholders who clarify requirements, validate completed functionality, and provide operational feedback.

Participant	Involvement in Kanban Development
Student Developer	Maintains the Kanban board, implements and tests work items, records progress, and updates task status.
Project Supervisor	Provides academic guidance, reviews progress, and advises on scope and research requirements.
Industry Supervisor	Provides industry and technical guidance and assists with implementation concerns.
HSAAS Stakeholders	Clarify operational requirements, review completed modules, and provide feedback for backlog refinement.

Proposed Kanban Workflow

Workflow Stage	Description
Backlog	Contains identified user stories, system requirements, technical tasks, defects, documentation tasks, and stakeholder feedback.
Ready	Contains prioritised work items with clear descriptions, acceptance criteria, and identified dependencies.
In Progress	Contains work items currently being implemented. An initial WIP limit of two work items will be used.
Review and Testing	Contains implemented work undergoing unit testing, integration testing, functional verification, and correction. An initial WIP limit of two work items will be used.
Stakeholder Validation	Contains completed functions requiring operational review or feedback from HSAAS representatives.
Done	Contains work items that have satisfied their acceptance criteria, passed the required testing, completed relevant documentation, and received required validation.

Development follows a pull-based workflow. A new work item is pulled from Ready into In Progress only when the In Progress column has available capacity. This prevents excessive parallel development and encourages completion of existing work before additional tasks are started.

Each work item will include a work item ID, module, user story or task description, priority, acceptance criteria, technical dependencies, test status, stakeholder comments, and completion status. Items that cannot progress because of missing information, technical problems, API availability, or pending stakeholder confirmation will be marked as Blocked.

Requirements collected through site visits, observation, interviews, and document review will be converted into work items in the Kanban backlog. New requirements, change requests, defects, and feedback may be added throughout development. Backlog items will be prioritised according to operational value, technical dependency, security and privacy risk, implementation complexity, and the available project timeline.

Testing is integrated into the Kanban workflow. Each implemented work item must pass the relevant unit and integration testing before it is moved to Stakeholder Validation or Done. Feedback obtained during stakeholder validation may return an item to In Progress or create a new backlog item for later improvement.

High Level Delivery Plan

Target Period	Planned Feature Group
June - July 2026	Project initiation, requirement analysis, technology evaluation, architecture design, database design, and Kanban board setup.
July - August 2026	QR-based visitor registration, four category-specific forms, backend APIs, initial database, and basic counter staff dashboard.
September - October 2026	Pass issuance and return, overdue detection, MRN validation, and blockchain audit trail integration.
November - December 2026	Real-time monitoring, historical analytics, administrator control panel, and blockchain transaction verification.

January 2027	End-to-end integration, system and performance testing, PDPA review, UAT, defect correction, and refinement.
February 2027	Pilot demonstration, staff training, documentation, final refinement, and FYP submission.

The target periods above provide a high-level roadmap rather than fixed sprint boundaries. Day-to-day development will be controlled through the Kanban board. Completed modules may be presented to HSAAS whenever they reach Stakeholder Validation. The Gantt Chart and Work Breakdown Structure in Chapter 8 describe the overall academic schedule, while the Kanban board manages the detailed flow of development work.

6.4 Data Collection Methods

Three complementary data collection approaches were used during and after the HSAAS industry site visit to elicit requirements for the QR-based Hospital Visitor and Pass Management System with Blockchain Technology. The first approach was to observe it directly. The developer witnessed the current manual process of visitor management at the HSAAS pass counter and learned about how visitors are currently registered, how the physical passes are issued, returned and how visitors information is recorded. The observation included the process of registering various classes of visitors such as Executive Visitor, Penjaga, Vendor and Contractor. It was also used to identify workflow inefficiencies, pain points in the customer journey and the order of interactions between the counter staff and the visitors.

Semi-structured interviews were a second method. Interviews were held with HSAAS counter staff and a representative from the Hospital management to obtain requirements that had not been obtained from observation. The interview questions centered on the ongoing concerns with registration, pass non-return issues, financial and operational impacts of lost passes, anticipated system features, privacy and institutional policy data handling concerns, and digital adoption concerns. All responses were consolidated and analysed to identify common themes. The identified requirements were then converted into user stories and work items in the Kanban backlog, where they were prioritised according to operational value and technical dependency.

Document review was the third method. Paper registration forms, samples of visitor logs and pass issuance records were reviewed to gain an understanding of the data currently collected by the hospital. This review was useful in determining the data fields needed for various categories of visitors and the type of data contained in MRN related information in Penjaga visitor registration. The results from the document review were used to develop digital registration forms for each category and to make sure that the new system would follow the existing HSAAS system's visitor pass process.

6.5 System Design (Architecture, Tools, Technologies)

This section describes the proposed system design for the QR-Based Hospital Visitor and Pass Management System with Blockchain Technology. The system is designed to support the digitalization of the existing visitor pass counter workflow at Hospital Sultan Abdul Aziz Shah (HSAAS). It focuses on QR-based visitor registration, physical pass lifecycle management, blockchain-based audit trail recording, real-time monitoring, and administrative control.

The system design consists of several key components, including the system architecture, user roles, functional modules, database design, and blockchain integration design. These components are designed to ensure that the proposed system is practical, secure, auditable, and suitable for the operational workflow of HSAAS.

6.5.1 System Architecture

The proposed system is based on multi-layer architecture, MVC-based application structure with integration of blockchain audit trail. This architecture is chosen due to the fact that the system comprises multiple functional components such as QR based visitor registration, visitor pass lifecycle management, real-time monitoring, analytics dashboard, database storage, and blockchain based audit trail recording. The proposed architecture allows for the system to be divided into different layers, which enhances maintainability, scalability, security, and system organization.

The system offers various interfaces for various kinds of users at the presentation layer. People utilize the system without having to make an account or sign in through a public QR enrollment interface. They scan a QR code at the HSAAS visitor pass counter, and are guided to a registration form on the HSAAS website. Visitors register through this form and choose a category (Executive Visitor, Penjaga, Vendor, Contractor). There are authenticated dashboard interfaces for

counter staff and administrators. Visitors submit their registrations on the counter, staff on the counter checks visitor information, issues physical passes, checks for pass returns and looks for overdue passes on the dashboard. The dashboard helps administrators to manage users, system settings, analytics, inventory of passes, and audit logs.

The controller layer processes requests for both public and staff interfaces that require authentication. Public controllers control visitor registration form submission and secured controllers control the counter staff and administrator activities including verifying visitors, issuing passes, recording pass return, visitor dashboard analytics, user management, audit trail review and system configuration. This layer is the first to be encountered between the user interface and application logic, ensuring that each request is directed to the correct service component.

The logic of the system is in the service layer. It handles visitor processing, form validation by category, passes issuing and return workflow, pass overdue detection, analytics data preparation, audit log generation and logging of blockchain transactions. For instance, when the counter staff hand out a physical visitor pass, the pass management service will update the status of the pass in the local database, record the time of issue, associate the visitor pass with the registration record and trigger the blockchain audit service to create a tamper-proof pass reference transaction. Likewise, if a pass is returned or is over due, then the service remembers the fact and updates the pass lifecycle status.

The data layer contains the operational data used by the system such as visitors, registration, visitor types, destination/ward references, pass records, pass transaction records, user accounts, audit logs and blockchain transaction references. In this way, only sensitive visitor information, which is not necessary to be recorded on the blockchain, is written in the local database, such as identification number, telephone number, patient MRN etc. This allows the system to keep a full log of the operations without sacrificing privacy and data protection.

Patient Related data has been created for Penjaga registration, for that it shall be validated using an external hospital Patient Information API. The proposed system is not a patient database and does not hold full patient medical records. The option is to enter the visitor MRN received from the ward form, which then gets sent to the hospital API to check. The API response can be used to check if the MRN is valid and if the patient is linked to the selected destination or ward. The local database will store only a limited amount of information about the visitor pass, including MRN, destination or

ward reference, verification status and verification time stamp, for managing and auditing visitor passes. In the proposed system sensitive medical information will not be stored.

The last layer would be the blockchain audit trail layer. This is NOT intended to replace the local database. Rather, it serves as an extra control point for specific visitor pass life cycle events. These events consist of pass issuance, overdue status generation, administrative changes, etc., to pass return. If this event does happen, the system will generate a transaction reference or “hash” using user-selected details of the transaction that are not sensitive information, such as the pass ID, transaction type, timestamp, staff ID, and visitor record reference. This transaction reference is sent to the blockchain testnet environment and from there the testnet environment returns the ID of the blockchain transaction which is stored in the local database. This way the system can check the authenticity of pass transactions without revealing any private data of the visitors, patients on the blockchain.

The detailed multi-layer system architecture diagram is provided in Appendix A to illustrate the relationship between the presentation layer, controller layer, service layer, data layer, external hospital API, and blockchain audit trail layer

6.5.2 User Role

The proposed system consists of three main user roles: Visitor, Counter Staff, and Administrator.

Visitor
Visitors are external users who scan the QR code at the pass counter and complete the digital registration form. They are required to select the appropriate visitor category, such as Executive Visitor, Penjaga, Vendor, or Contractor. Depending on the visitor category, different information may be required. For example, Penjaga visitors may need to provide patient MRN-related information based on the hospital’s existing workflow.
Counter Staff
Counter Staff are responsible for managing daily pass counter operations. They review incoming visitor registration records, verify visitor information, issue physical visitor passes, record pass returns, monitor overdue passes, and search visitor or pass records when required.

Administrator
Administrators are responsible for managing system-level functions. These include user account management, pass inventory management, report generation, analytics review, system configuration, audit log review, and blockchain transaction verification

The detailed use case diagram for the proposed system is provided in Appendix B to illustrate the interaction between Visitors, Counter Staff, Administrators, and external supporting components.

6.5.3 Functional Modules

A series of functional modules are divided into the proposed system. The first module is the Visitor Registration Module. This module enables walk-in visitors to sign in by scanning a QR code that is presented at the hospital pass counter. Visitors scan the QR code next, and then proceed to a browser-based visitor registration form to choose from the available categories and fill in the information. The system allows for various types of visitors: Executive Visitor, Penjaga, Vendor and Contractor. Different registration fields may be required depending on the purpose of the category and the level of access it requires. This module minimises reliance on paper based forms, and enhances the consistency, legibility and searchability of visitors.

There is a second module, the Visitor Pass Management Module. This module lets counter staff issue verified visitors a physical pass. Passes are tied to a visitor record and given a pass status for each pass. The system keeps track of key data pertaining to the pass, such as pass ID, pass issuance date, staff assigned to pass, visitor category and the destination. The visitor will be removed from the hospital and the pass return will be entered in the system. The status of the pass, and the time when it was returned, is then updated in the system. When a pass is not returned within the allotted time period the system issues an alert for the pass to be returned as overdue. This aids staff to identify visitors who haven't returned their pass and call them on the number that they have registered on.

The third module is Blockchain Audit Trail Module. The sole purpose of this module is to log a subset of the pass lifecycle events on the Sui Testnet to enhance the integrity, traceability and accountability of pass transactions. If a pass is issued, returned, marked as overdue or

changed by an administrator, then a transaction event is created. Then a hash or transaction reference is added on the blockchain and the blockchain transaction ID is added to the local database and is associated with the corresponding pass transaction record. Visitor data is not directly recorded on the blockchain; sensitive visitor data (such as identification number, phone number, MRN) is not directly recorded on the blockchain. This design allows the system to check if a pass transaction record has been altered after creation, yet keeps privacy risks to a minimum.

Monitoring and Analytics Dashboard Module is the fourth module. This module offers operational visibility of the counter in real time to staff and administrators. The dashboard includes details on active pass holders, overdue passes, visitors by destination, visitor category trends, and pass utilisation status. This module can be used for operational decision making and enable the hospital to determine the time of peak visitor and monitor pass usage, detect overdue passes and analyse visitor registration patterns. The dashboard will only show registered visitor pass activities, and not monitor all Hospital visitors.

The fifth module is the Administration Module. This module enables the following access to user accounts, configuration of system settings, pass inventory management, audit log viewing, and reporting functions. The blockchain transaction ID for each pass lifecycle event can be used to check blockchain transaction records, as well. This module enables the maintenance of systems, administrative control and accountability for visitor pass management operations.

6.5.4 Database Design

The system uses a relational database to store operational data required for visitor registration, pass management, user management, and reporting. The main database entities include User, Visitor, Visitor Registration, Visitor Category, Ward, Pass, Pass Transaction, Blockchain Transaction, and Audit Log.

The Visitor entity stores basic visitor details such as name, identification number, and phone number. The Visitor Registration entity stores each registration record, including visitor category, destination ward, registration date and time, and MRN-related information where applicable. The Pass entity stores physical pass details, including pass ID and pass status. The Pass Transaction entity records pass issuance, return, overdue, and administrative modification

events. The Blockchain Transaction entity stores blockchain transaction references linked to the corresponding pass transaction.

The detailed entity relationship diagram (ERD) is provided in Appendix C to show the database structure, main entities, attributes, and relationships used to support visitor registration, pass lifecycle management, audit logging, and blockchain transaction references.

6.5.5 Blockchain Integration Design

The blockchain integration is supposed to act in an audit trail role and not as the main data storage layer. Visitor records, pass records and operational data are still stored primarily within the local database. Only specific evidence of transactions are recorded on blockchain for pass lifecycle events.

When a pass transaction happens, the blockchain integration process starts. For instance, if a counter staff member gives a pass, the event is logged in the local database backend. The system then creates a transaction hash (or reference) based on certain non-sensitive transaction information including pass ID, transaction type, date/time, staff ID, and visitor record reference. This is a reference to a transaction that is given to the blockchain Testnet. Once the blockchain transaction is confirmed, the blockchain transaction ID returned will be saved locally.

Such a design enables the system to confirm the integrity of the pass transaction records while not revealing any sensitive information of the visitors on the blockchain. If there is a pass transaction record later, the stored database record can be compared with the blockchain transaction reference to find any discrepancies.

6.6 Software Verification and Validation

Software verification and validation will be carried out to ensure that the proposed QR-Based Hospital Visitor and Pass Management System with Blockchain Technology fulfils the requirements of HSAAS. Verification determines whether the system is constructed according to its design and functional requirements, while validation determines whether the completed functions satisfy stakeholder expectations and actual hospital operations.

Verification and validation will be performed continuously as work items move through the Kanban workflow and will be fully executed during integration, system testing, and user acceptance testing.

Unit Testing
For each development work item, the developer will test the relevant function, API endpoint, user-interface component, or data-processing logic before the item is moved from In Progress to Review and Testing. Tests will cover valid, boundary, and invalid data, including registration validation, pass lifecycle transitions, overdue duration calculation, and blockchain hash calculation.
Integration Testing
Integration testing confirms the functions of the various system modules work together properly from QR form submission to dashboard notification, from pass issuing event recording to blockchain transaction commitment and ID storage. Integration tests are used to test the blockchain integration pipeline, ensuring that all passed transaction events create a valid blockchain transaction, and that transaction ID is properly stored in the local database.
System Testing
System testing involves running end-to-end workflows and all four types of visitors to ensure that all the flows and acceptance criteria in the original user story are followed, including: all four visitor category registration flows with valid and invalid data; the full pass lifecycle from issuance to return; overdue alerts and contact display; and role-based access enforcement for all three user roles for the pass system.
User Acceptance Testing (UAT)
UAT is performed at HSAAS with participation of counter staff and administrators, who are operating the live system in a supervised environment. UAT tests the entire system workflow to ensure visitors are registered with MRN entry, passes are issued, visitors return, and the

overdue scenario is simulated, while keeping you monitored on dashboard. Input for final system refinement will come from feedback gathered during UAT.

6.7 Evaluation Methods

The four dimensions of the QR-based Hospital Visitor and Pass Management System with Blockchain Technology will be used to perform the usability evaluation to determine the expediency of the system in fulfilling the goals and achieving measurable operational improvements in HSAAS. Staff working counters who take part in UAT will be required to fill out the System Usability Scale (SUS) form. SUS ranges from 0 to 100 with anything above 68 being considered good usability. The SUS is a single composite usability score and offers sub-dimension analysis of learnability and usability. SUS is defined as the minimum acceptable score in this system as the acceptable usability for the intended counter staff population.

Registration Efficiency

The processing time for the registration operation (from QR scan to counter staff dashboard processing validation and issuance of the pass) will be measured during UAT by a stopwatch for a set of test registration scenarios for all the four categories of visitors. The aim is to produce an average processing time of less than 5 minutes per visitor, which is much longer for the manual process as observed on site. .

Pass Management Effectiveness

The accuracy for detecting an overdue will be tested using a set of overdue scenarios (passes that aren't returned after selected maximum time) and confirming that all of the pass transaction events tested result in an accurate alert being sent, with the visitor's contact number displayed. The transitions for each pass will be checked with a collection of test cases that handle every possible valid transition of the pass (Available → Issued → Available; Available → Issued → Overdue → Available).

Blockchain Audit Trail Integrity

Evaluating the effectiveness of the Blockchain integration will be done by verifying that 100 percent of pass transaction events will result in a corresponding blockchain transaction with a stored transaction ID; confirming that the blockchain verifier will correctly detect the tampered local records in the controlled tampering test; and measuring mean blockchain transaction latency to ensure that it does not materially affect counter staff workflow.

Dashboard and Analytics Accuracy

It will be checked during the UAT phase by comparing the metrics displayed on the real time dashboard with the underlying state of the database. History analytics will be checked by ensuring that the analytics produced are accurate representations of the data put into the system during system testing.

6.8 Summary

The methodology adopted for the development of the QR-Based Hospital Visitor and Pass Management System with Blockchain Technology consists of Design and Development Research (DDR) as the research design and Agile Kanban as the software development methodology. Requirements are identified through site visits, direct observation, semi-structured interviews, and document review at HSAAS. Development work is visualised and continuously managed through the stages of Backlog, Ready, In Progress, Review and Testing, Stakeholder Validation, and Done. Work-in-progress limits and a pull-based workflow are used to maintain development focus. Unit, integration, system, and user acceptance testing are integrated into the workflow, while usability, registration efficiency, pass management effectiveness, blockchain audit trail integrity, and dashboard accuracy are evaluated against the project objectives.

7. Return of Investment (ROI)

This section analyzes the expected costs and benefits associated with the development and deployment of the QR-based Hospital Visitor and Pass Management System with Blockchain Technology at HSAAS, providing a structured assessment of the return on investment the hospital can expect from adopting the proposed system.

7.1 Estimated Cost

As this system is developed as a Final Year Project (FYP) by a student developer, the development labour cost is absorbed within the academic project rather than charged as a commercial development expense. The primary cost categories for HSAAS are as follows:

Component	Descriptions
Infrastructure and Hosting Cost	The system needs a server for backend API, database and blockchain node. Cloud hosting would be better choices because of its price. For small to medium size institutional use, it is estimated that lightweight cloud hosting on a small number of resources will cost about RM 50–150 per month, depending on the cloud provider and the resource configuration used.
Coding Supporting Tools (ChatGPT/Claude)	The coding supporting tools used to assist developers in coding, debugging and documentation while improving development efficiency. A ChatGPT Plus account is RM 99.90 /month while a Claude Pro account is USD 17 /month
Blockchain Network Setup Cost	Blockchain Sui testnet usage for development and verification purposes will not bring any additional charges but a live mainnet token price would be 0.76 USD, the transaction fee depends on computation gas.
Staff Training Cost	There are 2 to 3 hours estimated to be spent on training sessions with counter staff and administrators. The training expense is mostly a cost of staff time, and a small cost for a pilot deployment.
Maintenance Cost	Post-handover maintenance is estimated to be 2 to 4 hours per month for system monitoring, minor updates and data backup verifications, which are easily handled by the hospital IT department.

7.2 Expected Benefits

The expected benefits of the proposed system can be divided into tangible and intangible benefits. Tangible benefits refer to benefits that can be observed, measured or estimated directly while intangible benefits refer to benefits that may not be directly measured in financial terms but important for organizational value.

7.2.1 Tangible Benefits

The following are the definite advantages of the proposed system to the HSAAS in reducing pass loss, increase in the efficiency of registration, lessen workload of the counter staff and support the management of pass inventory. The automated overdue alert function can also help minimize the loss of visitor passes and replacement expenses by alerting counter personnel when a visitor pass has not been returned within the set time limit. This allows staff to call the visitor on the registered telephone number and be able to follow up on the unreturned pass more efficiently. Each Physical Pass costs about RM30 and each time the pass is lost, there is a corresponding financial loss which can be avoided if improved pass recovery is achieved.

Secondly, the system can cut the time spent by the visitors in registering by switching the handwritten forms and logbook entries with QR-based digital registration. To submit the information, visitors are asked to scan the QR code on the pass counter and fill out the information on a browser based form from their own device. This minimises manual writing, makes data easier to read and speeds up the process from arrival of the visitor to the counter employee verifying his or her identity and issuing the pass.

Third, the system minimizes manual effort required by the counter staff by offering a digital dashboard to review registration, issue passes, record pass returns, notify of overdue passes, and track visitors. The manual review of logbooks and pass status of visitors has been eliminated from the counter routine. This enables staff to better organise and efficiently manage visitor pass activity.

Fourth, the system optimizes pass inventory management with the ability to keep up with the information about the availability and condition of passes in real time. The system is capable of displaying the status of a pass (Available, Issued, Overdue, Lost, or Disabled). This decreases the need for manual pass counting and helps the hospital to control their physical pass inventory better.

7.2.2 Intangible Benefits

The proposed system will bring to HSAAS and also several intangible benefits. A major advantage is increased accountability of the institutions. The system establishes digital audit logs and records specific critical moments in the pass lives as references on the blockchain to build up the authenticity of pass issuance, return, overdue status, and pass administration modification. This enhances accountability when it comes to visitor access management and increases confidence in pass transaction reliability.

The system also enhances the visitors' experience by offering a quicker, paperless, and hassle-free registration process. Those who must register will use their own device to fill out the registration form by scanning a QR code; no application is to be installed, nor a system account created. This will decrease the hassle of handwritten paperwork in addition to possibly minimizing the period spent at the pass counter.

Another non-financial advantage is the ability to make better operational decisions based on data. These real-time dashboard and historical analytics allow hospital administrators to access the categories of visitors, distribution of destination points, number of active passes, overdue passes, pass utilization, and historical trends of visitors. They can be used to make more informed decisions on counter staffing, enforcement of visitor policy, ward/destination monitoring, and/or pass inventory planning.

The proposed system also enhances the data governance and compliance position of the hospital. The system provides better handling of visitor data than paper-based records due to structured digital records, role-based access control, audit logging, and controlled storage of the visitor's information. The system allows for sensitive personal information and patient-related information to be stored in the local database, which can be used to manage personal information in a privacy-aware manner while enhancing accountability and traceability on the blockchain.

7.3 Value Contribution to Stakeholder

The proposed system creates value to several stakeholder groups, the visitors, counter staff, administration, and management of the hospital. The system provides a problem-free and quick sign-in process for visitors. Visitors are required to register who will have to walk in can scan the QR code and submit the information via a browser without downloading any application. The system can also be used for a more streamlined registration process for Penjaga

visitors, depending on the availability of integrating with the hospital information system, in order to provide a more effective MRN-related registration process.

The system streamlines the manual processes on counter staff, eliminating the need for paper logbooks and manual pass tracking by providing a dashboard-driven workflow. The submitted registrations can be viewed on the counter and information on visitors can be verified, physical passes issued, returned passes recorded, active and overdue passes monitored, and contact information for visitors checked and followed up. This assists with the daily counter operations as well as enables staff to respond faster to unreturned pass cases.

Administrators have access to system-level management features including pass management, pass inventory management, system configuration, report generation, system analysis, review of analytics, monitoring of audit logs, and blockchain transaction verification. These functions facilitate more effective management of the activity of the visitor passes and enhance accountability for system activities.

The system offers improved operational visibility to the hospital management by giving analytics about the types of visitors, their destinations, numbers of active passes, expired passes, and the trend of visitors in the past. The data produced by the system can help make more proactive and data-driven decisions on counter staffing, pass monitoring, destination monitoring, and pass inventory planning. In general, the proposed system helps to improve the management process of the visitor passes at HSAAS in a more structured, traceable and accountable way. structured, traceable and accountable way.

8. Project Planning

8.1 Work Breakdown Structure (WBS)

The Work Breakdown Structure divides the project into major workstreams and supporting activities aligned with the Agile Kanban methodology described in Section 6.3. The workstreams provide high-level planning categories, while individual activities are continuously managed as work items through the Kanban board. The WBS forms the basis of the Gantt Chart in Section 8.2 and links project activities, target periods, deliverables, and system objectives.

ID	Workstream	Task	Deliverable	Target Period
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1.0	Project Management			Ongoing
1.1		Project planning and scheduling	Project plan	Jun 2026
1.2		Stakeholder communication	Meeting records	Ongoing
1.3		Progress monitoring and reporting	Status reports	Ongoing
2.0	Initiation and Kanban Setup			Jun-Jul 2026
2.1		Site visit synthesis and documentation	Requirements document	Jun 2026
2.2		Stakeholder interviews and workflow analysis	Interview notes	Jun 2026
2.3		Technology stack evaluation and selection	Technical specification	Jun-Jul 2026
2.4		System architecture and database design	Architecture diagram	Jun-Jul 2026
2.5		Kanban backlog and board setup with HSAAS	Initial Kanban board	Jun-Jul 2026
3.0	QR Registration			Jul-Aug 2026
3.1		QR code generation and counter display	QR display	Jul 2026
3.2		Visitor registration forms for four categories	Working digital forms	Jul-Aug 2026
3.3		Backend API and database implementation	API endpoints and schema	Jul-Aug 2026
3.4		Basic counter staff dashboard	Dashboard version 1	Aug 2026
3.5		Review, testing, and stakeholder validation	Validation record	Aug 2026
4.0	Pass Management and Blockchain			Sep-Oct 2026

4.1		Pass issuance and return module	Pass management module	Sep 2026
4.2		Overdue detection and alert system	Alert functionality	Sep-Oct 2026
4.3		MRN validation integration for Penjaga	MRN validation	Oct 2026
4.4		Blockchain audit trail integration	Blockchain module	Sep-Oct 2026
4.5		Review, testing, and stakeholder validation	Validation record	Oct 2026
5.0	Analytics and Administration			Nov-Dec 2026
5.1		Real-time monitoring dashboard	Dashboard version 2	Nov 2026
5.2		Historical analytics and reporting	Analytics module	Nov-Dec 2026
5.3		Administrator control panel	Admin panel	Dec 2026
5.4		Blockchain transaction verifier	Verifier interface	Dec 2026
5.5		Review, testing, and stakeholder validation	Validation record	Dec 2026
6.0	Integration, UAT, and Refinement			Jan 2027
6.1		End-to-end integration and system testing	Test report	Jan 2027
6.2		User Acceptance Testing at HSAAS	UAT record	Jan 2027
6.3		Blockchain integrity verification	Verification report	Jan 2027
6.4		PDPA and data-handling review	Compliance checklist	Jan 2027
6.5		Performance testing and defect refinement	Performance report	Jan 2027

7.0	Deployment and Handover			Feb 2027
7.1		Pilot demonstration at HSAAS counter	Operational prototype	Feb 2027
7.2		Staff training and handover	Training materials	Feb 2027
7.3		User manual and administrator guide	Documentation	Feb 2027
7.4		Final report and FYP submission	Final report	Feb 2027

Note: The target periods are high-level planning windows. Individual work items are managed continuously through the Kanban board and may be reprioritised when technical dependencies or stakeholder feedback require changes.

8.2 Gantt Chart

Table 8.2 presents the high-level project roadmap covering June 2026 to February 2027. The Gantt Chart provides planned target periods and major milestones, while detailed development activities are continuously managed through the Kanban board. The periods represent planning targets rather than fixed sprint boundaries.

Workstream	Jun 26	Jul 26	Aug 26	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Feb 27
Project Initiation and Kanban Setup	X	X							
Architecture approval milestone		M							
QR Registration		X	X						
QR module stakeholder validation			M						
Pass Management and Blockchain				X	X				

Pass and blockchain validation					M				
Analytics and Administration						X	X		
Analytics and admin validation							M		
Integration, UAT, and Refinement								X	
UAT approval milestone								M	
Deployment and Documentation									X
Final FYP submission milestone									M

From June to July 2026, the project focuses on requirements, technology selection, architecture, database design, and Kanban setup. QR registration is targeted for July to August, followed by pass lifecycle management, MRN validation, overdue alerts, and blockchain integration from September to October. Analytics and administration are targeted for November to December. Integration, UAT, security and PDPA review, and refinement are planned for January 2027, followed by deployment preparation, documentation, training, and final submission in February 2027. Individual work items will continue to move through the Kanban workflow according to priority, dependency, and available development capacity.

9. Conclusion

9.1 Summary of Proposal

The QR Based Hospital Visitor Pass Management System using Blockchain with respect to Hospital Sultan Abdul Aziz Shah (HSAAS) is proposed to develop. The proposed system is aimed at digitizing the current manual visitor pass counter process, including QR-based visitor registration, the creation of forms specific to the visitors' categories, management of visitor pass'

lifecycle, support for the verification of MRNs, the recording of the audit trail through blockchain and the implementation of an analytics dashboard.

The project is driven by the operational difficulties found at HSAAS, such as paper registration of visitors, manual tracking of visitors' passes, finding unreturned passes, lack of operational visibility and lack of traceability of the audit process. This system overcomes these problems by offering a centralized digital platform for visitor registration, pass issuance, pass return, overdue alert monitoring, and reporting.

The following are the major roles of users for the proposed system: Visitor, Counter Staff and Administrator. Users can scan QR codes and fill in the registration forms without registration. The counter staff can check registrations, issue passes, record pass returns, and keep track of overdue passes. Users can add and remove users, set system parameters, monitor analytics, create reports, and check blockchain transaction references.

The system is built on a multi-layer architecture, featuring an MVC (Model-View-Controller) approach to application structure and integration of blockchain auditing trail. Certain non-sensitive pass lifecycle transaction references are recorded on the blockchain audit trail layer with sensitive visitor and patient related information being stored in the local database. This design helps to ensure operational efficiency and enhance the integrity and traceability of critical pass transaction information.

The project will be developed using the Agile Kanban methodology, enabling development work to be visualised, prioritised, tested, and progressively completed through a continuous workflow. Stakeholder feedback will be incorporated into the Kanban backlog, allowing the system to be refined according to the operational requirements of HSAAS. Verification, validation, and evaluation activities will assess the functionality, usability, performance, security, and suitability of the system for hospital operations.

9.2 Expected Contribution

The proposed system will make contributions to HSAAS and the field of Healthcare information system development in several aspects.

Firstly, the system helps to achieve the hospital's operational efficiency by minimizing the reliance on paper visitor registration and manual pass management. QR registration will enable visitors to submit their data digitally and the staff member on the counter can see it on their dashboard to check.

Secondly, the system helps to ensure visitors have accountability by offering a structured pass lifecycle management. Every physical pass can be tracked by their current status, such as available, issued, overdue, lost, disabled etc. Pass transaction events are recorded for pass issuance, pass return, pass overdue generation and administrative pass modifications.

Third, the system helps to ensure audit trail integrity by incorporating blockchain as an immutable and secure verification layer for key pass lifecycle events. The blockchain is not a substitute for the local database, but an extra way to check if key pass transaction records have been altered.

Fourth, the system helps with operational visibility by providing analytics regarding the categories of visitors, active passes, distribution of visitors by destination, overdue passes, pass use, and historical visitor trends. Such analytics can help hospital management track and analyze visitor pass operations, enabling data-driven decisions.

Fifth, the system helps to enhance data governance by implementing role-based access control, audit trails, data structuring, and controlled management of visitor and MRN data. The system does not store sensitive information of visitors/patients directly on the blockchain, thus minimising privacy and data protection risks.

Lastly, the project is a working software engineering prototype that includes QR-based registration, pass lifecycle management, blockchain audit trail verification and analytics dashboard capabilities within one tool for hospital visitor pass management. The system is designed on the basis of the actual needs observed in the operation of the hospital HSAAS, and can be used as a basis for further development, deployment or integration into the general smart hospital project.

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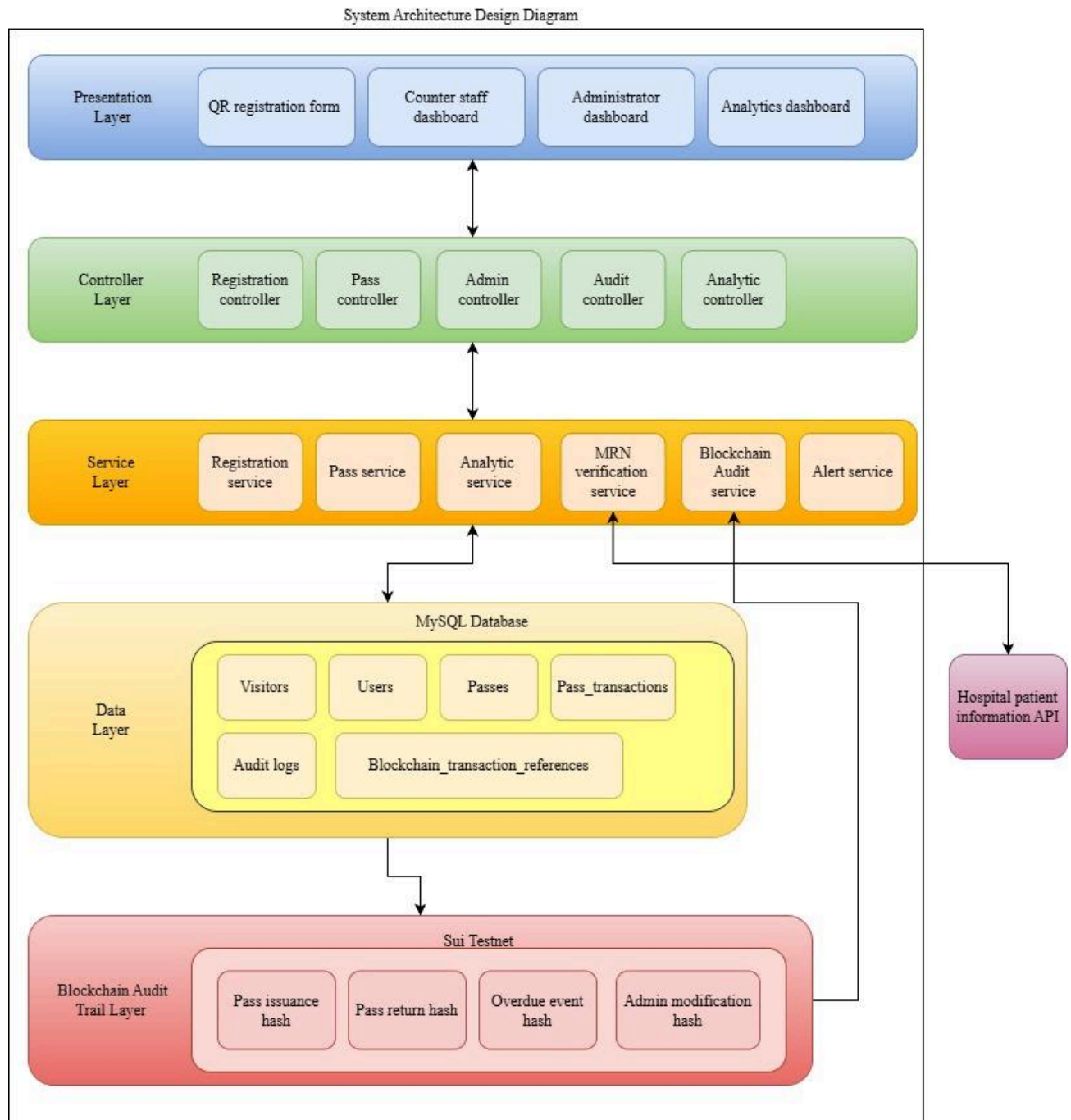
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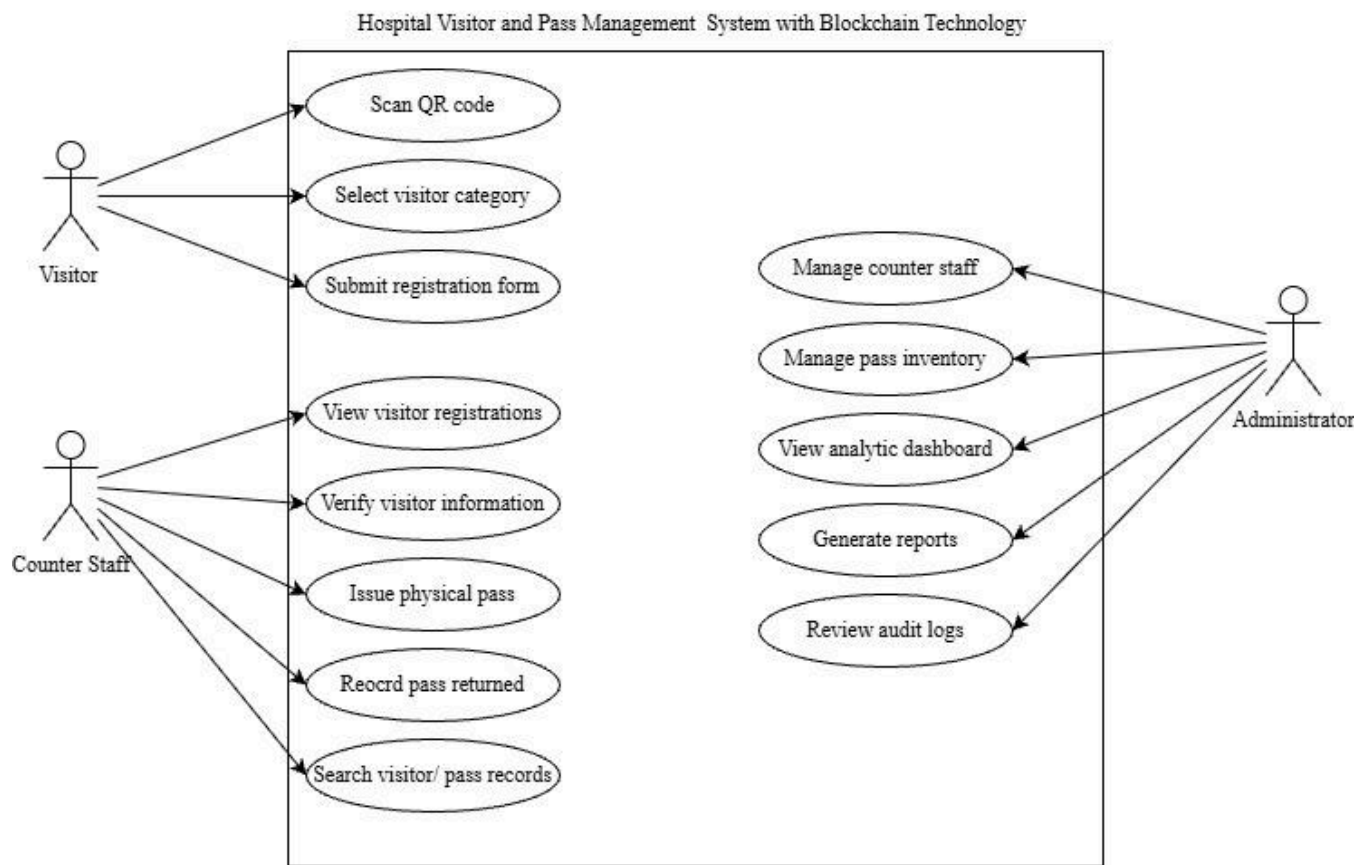
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11. Appendices



Appendix A



Appendix B

